CAPSTONE PROJECT | E-COMMERCE ANALYTICS

# ShopSphere Conversion Intelligence

*Four Teams. One Unanswered Question.*

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# Agenda

# Four Teams, One Unanswered Question

ShopSphere has increased marketing spend across every channel. Traffic is growing — but CAC is rising, ROAS is declining, and four teams are each seeing a different symptom of the same undiagnosed problem.

| Marketing | Finance | Customer Experience | Product |
|---|---|---|---|
| High clicks, few conversions | Rising CAC, declining ROAS | Inconsistent engagement across segments | Unexplained cart-abandonment variation |

**One question: What truly drives conversion?**

# How We Approached It

Before answering "what drives conversion," we had to define the question precisely — and be honest about how we'd know if we were wrong.

### What We're Solving For

Identify the customer, behavioral, and campaign drivers of conversion — and use them to improve targeting efficiency and marketing ROI.

### What "Solved" Looks Like

An interpretable propensity model; drivers ranked by actual contribution, not guesswork; every claim tested, not assumed.

### How We'd Know If We're Fooling Ourselves

A dual-pipeline design (Model A vs. Model B) built specifically to catch the analysis lying to us before we trust it.

# Why These Techniques

*The data's shape decided the toolkit — not the reverse*

Part of not fooling ourselves was resisting the temptation to reach for every advanced technique available. Here's what we used, and — just as importantly — what we deliberately didn't.

## USED

### Random Forest + Logistic Regression

Tabular, session-aggregated data with mixed feature types — exactly what tree ensembles handle natively

### SHAP

Required to trace every recommendation back to a specific, auditable driver

### Autoencoder

The one deep-learning tool that matches an unsupervised anomaly-detection task on numeric features

## NOT USED

### CNN

Built for spatial/image data — nothing in this dataset is visual

### RNN / LSTM

Needs per-click sequential logs; this data is pre-aggregated per session

### GAN

Built to generate synthetic samples — not this project's objective

# A Complication

*Before we could answer the question, the data asked one of its own*

**83.8%** vs. **1–5%**

observed conversion rate in the data

what unfiltered e-commerce traffic typically looks like

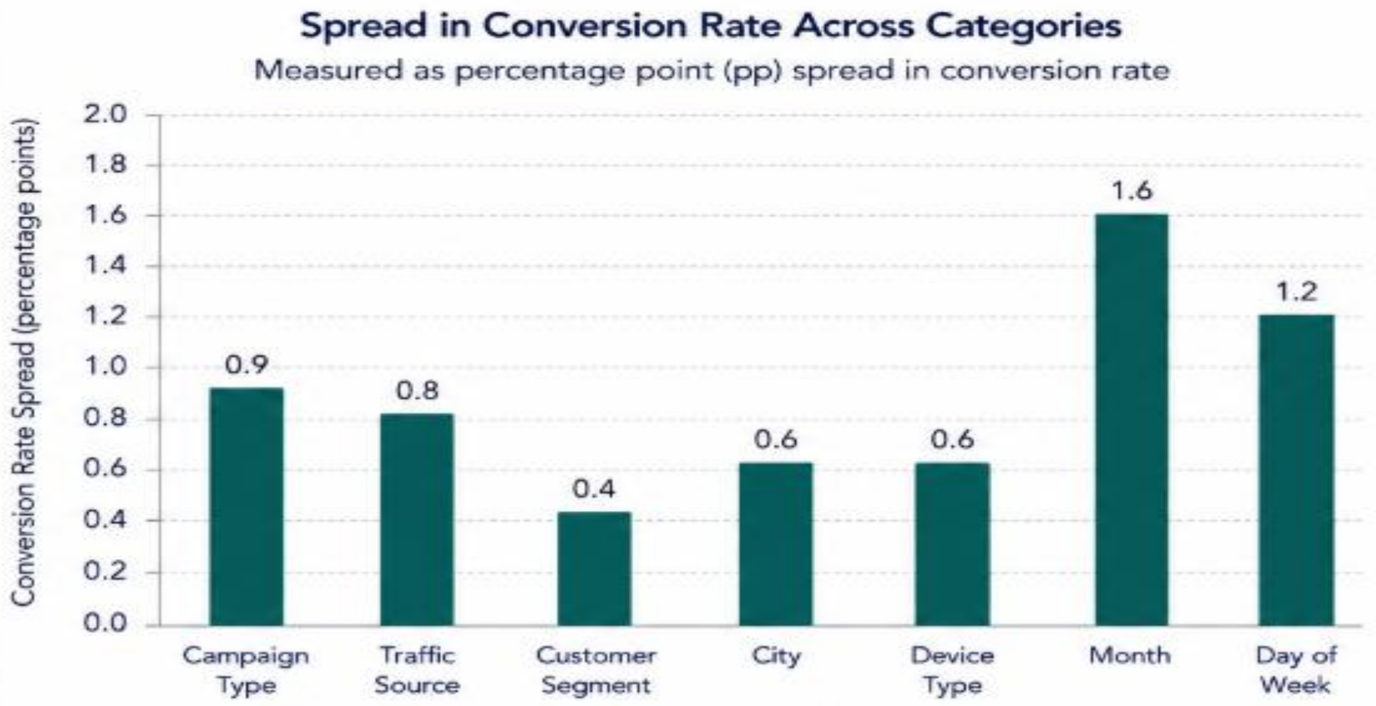
A gap this size isn't a rounding error. Either this data means something different than it appears to — or it isn't behaving like real customer traffic at all.

**We couldn't responsibly answer "what drives conversion" until we knew which.**

We'll return to that. For now — back to the original question: what actually drives conversion?

# Ruling Out the Obvious

*The natural suspects — channel, campaign, demographics — turned out not to be it*




Sample sizes are well-balanced (~10,000 sessions each) across every category. (Actual conversion rates across categories range from **82.9% to 84.6%**.)

## Under 1.6

**percentage points**

of spread across every category tested — campaign type, channel, segment, city, device, month, day of week — with well-balanced sample sizes (~10,000 sessions each), ruling out small-sample noise as the explanation.

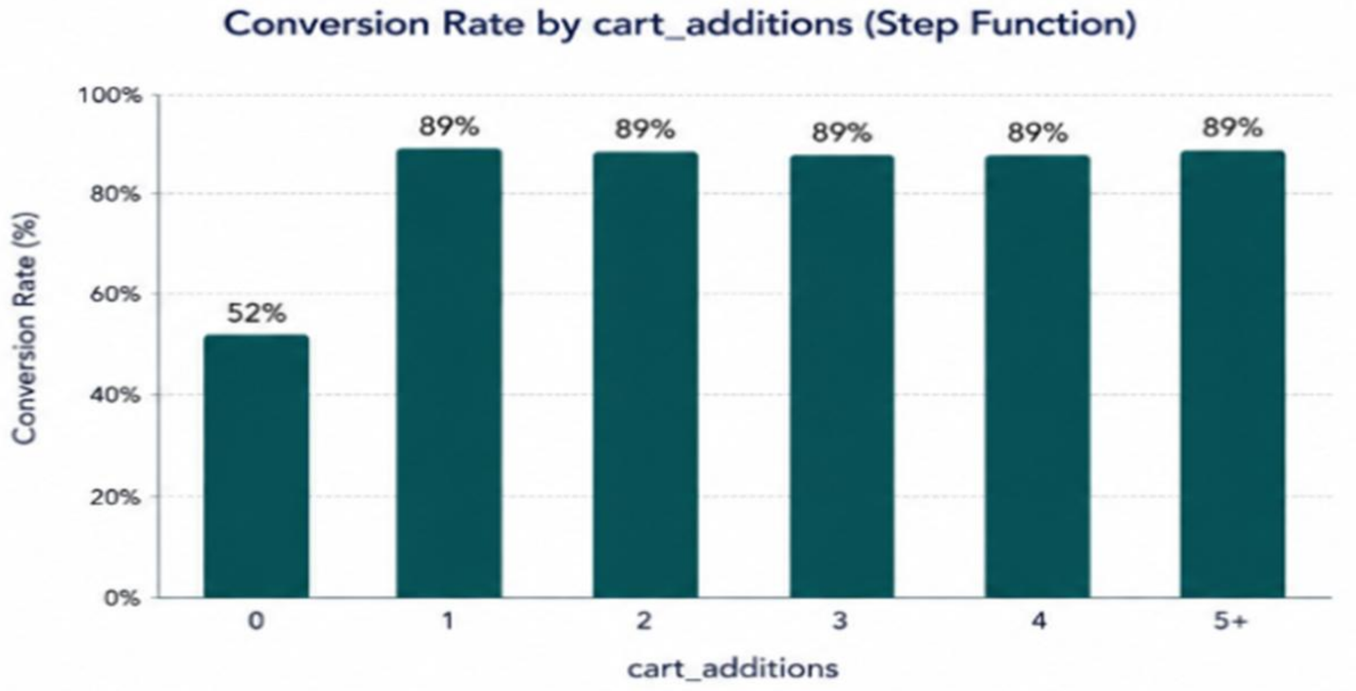
**First conclusion:**

***If it's not who they are or how they arrived, it has to be something they do.*** *(next)*

Not who they are, or how they arrived. What they **do** once they're on the site.

# The Real Driver

*In-session behavior — not acquisition — is what separates converters from everyone else*




Pattern holds across all major channels with consistent sample sizes (~10,000 sessions each).

## A step, not a slope

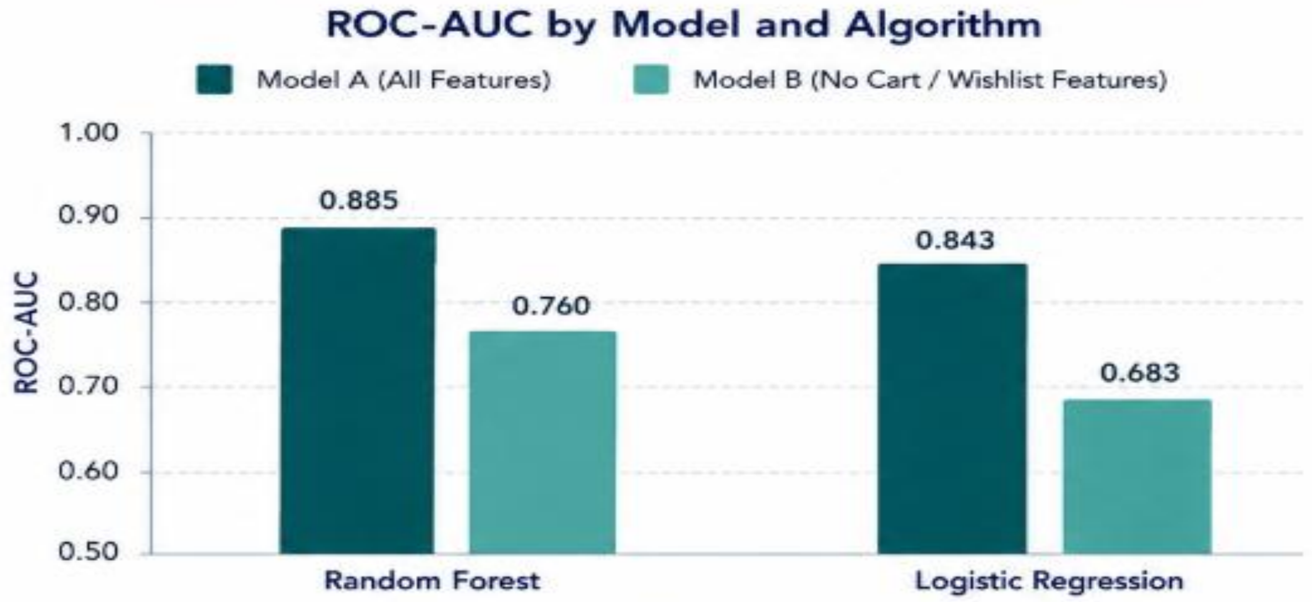
52% → 89% the moment **ANY** cart activity occurs, then flat regardless of quantity. Same shape holds — with graded severity — for wishlist_additions and discount_viewed.

> *But a jump this clean, this uniform across every channel, raised a different question: is this a genuine signal — or too good to be true?*

So we built a test designed to catch ourselves being wrong.

# Stress-Testing the Finding

*Model A sees everything. Model B is deliberately blind to cart and wishlist activity.*
*If the gap between them is small, the signal is real.*




Higher is better. Model A includes cart_additions and wishlist_additions. Model B excludes both.

## First, a quieter finding:

Random Forest beats Logistic Regression on every comparison — confirming the modeling approach itself holds up.

## Then, the answer we were bracing for:

The gap between Model A and Model B doesn't shrink — it grows under the simpler model. That's not what a fluke looks like. That's what a real, structural property of the data looks like.

**The signal is real.** But "real" and "fully understood" aren't the same thing — that reckoning comes later.

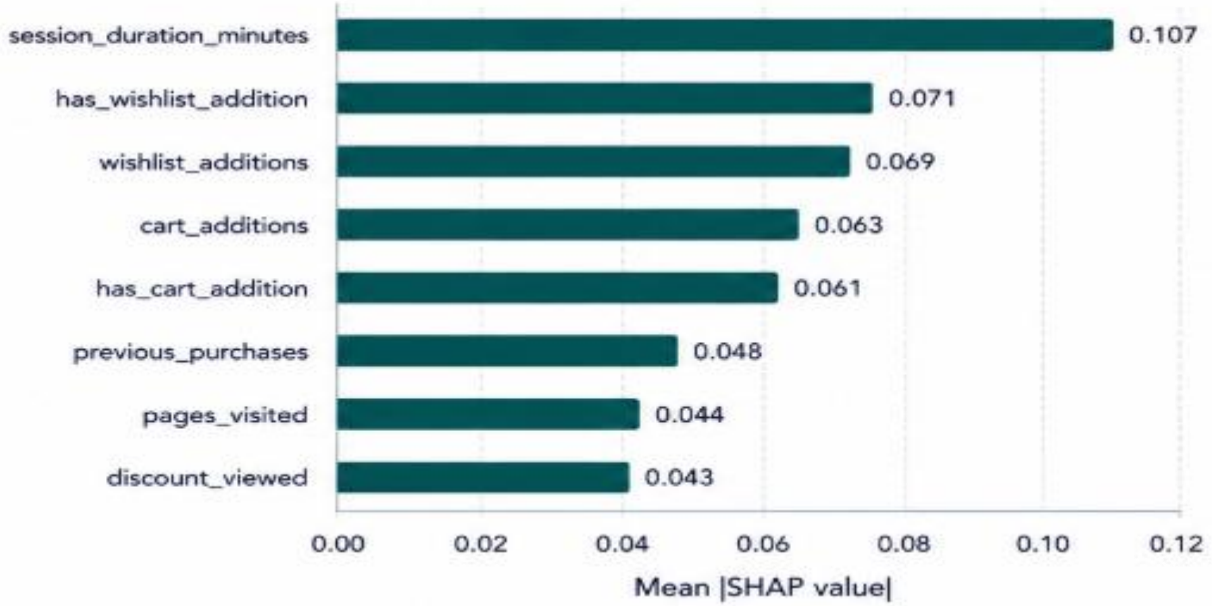
“Real” needed a number. So we asked the model to show its work.

# Precisely What Matters

*SHAP doesn’t just rank features — it tells you exactly how much weight each one carries*

### Top 8 Features by SHAP Importance (Model A)

Mean |SHAP value| (Global)




SHAP values represent the average absolute contribution of each feature to the model’s predictions. Higher means more important.

## 49.1%

of everything the model knows sits in just 5 of 51 features — cart and wishlist activity, concentrated to a degree that demands attention, not comfort.

### But the top feature wasn’t cart activity at all.

session_duration_minutes ranked #1 — despite showing almost no relationship with conversion in our earlier, simpler checks.

### A contradiction like that doesn’t get explained away. It gets investigated.

The contradiction had an answer: 100 sessions, all sharing one impossible number.

# An Honest Miss

*session_duration_minutes = 1000.0, in exactly 100 rows — confirmed as an injected sentinel value, not real behavior. We built an* **autoencoder** *to find things like it automatically.*

### Anomaly Detection Results

| Evaluation | Result |
|---|---|
| Reconstruction error: anomalies vs. normal | 0.795 vs 0.656 (+21%) |
| Statistical significance (t-test) | $p = 5.66 \times 10^{-10}$ |
| Top-100 flagged rows: true anomalies caught | 0 out of 100 |
| ROC-AUC (anomaly ranking quality) | 0.660 (real, but weak) |

## The test worked. The detector didn't.

The difference between anomalous and normal sessions is real ($p < 0.0001$) — but the detector couldn't act on it. 0 of the 100 known anomalies made the top-100 flagged list.

*We're showing you this because a deck that only shows what worked isn't showing you the whole analysis.*

**Which brings us back to the question we set aside at the start.**

Two questions have been sitting with us since the start. Time to answer both.

# Answering the Original Question

## On the 83.8%:

We can't fully resolve it — but the evidence points somewhere specific. Five separate, independent findings across this analysis (the base rate, channel uniformity, zero multicollinearity, the algorithm-independent leakage gap, and the sentinel-value outliers) all point the same direction: **this dataset behaves like it was constructed, not captured.** That's not a flaw in the analysis — it's the analysis working correctly.

1. Base rate: 83.8% overall conversion
2. Channel uniformity: <1.6 pp spread
3. Zero multicollinearity: no redundancy
4. Leakage gap persists (and widens) across models
5. Sentinel-value outliers: 100 sessions at 1000.0

## On what drives conversion:

Not channel. Not campaign.
Not who someone is or how they arrived.
**What they do in the first few minutes on the site — cart activity, wishlist activity, discount visibility — is the entire signal that matters.**

Cart Activity | Wishlist Activity | Discount Visibility

**No recommendation here has been deployed.**
**No before/after outcome has been measured.**
*This is diagnostic clarity — not yet a measured result.*

**But diagnostic clarity has value of its own:**
**it stopped a plausible, expensive mistake before it happened.**
ShopSphere was set up to keep optimizing channel mix — a lever this analysis shows doesn't move the outcome. Knowing that before spending more on it is the win.

**So — what should ShopSphere actually do?**

In order of confidence — from what the evidence demands to what still needs proving.

# What ShopSphere Should Do Next

| | | | |
|---|---|---|---|
| 01 | Resolve the complication first | Everything below depends on it. Validate what conversion_flag actually measures against real ShopSphere analytics before acting on anything else here. | CONFIDENCE Very High |
| 02 | Stop optimizing channel mix | No channel outperforms another. The lever that matters is what happens after arrival, not which channel brought the visitor. | CONFIDENCE High |
| 03 | Use early-funnel signals for real-time intervention | Model B — the version blind to cart activity — still predicts meaningfully. Usable today for in-session nudges before drop-off. | CONFIDENCE Medium–High |
| 04 | Retire click volume as a campaign KPI | ad_clicks and email_clicks show ~zero correlation with conversion. | CONFIDENCE Medium |
| 05 | Treat cart/wishlist targeting as provisional | Re-run the leakage test on validated real data before committing spend to it. | CONFIDENCE Medium |

**These are the actions the evidence supports.** But evidence has **boundaries** — and ours deserve to be **explicit.**

Five findings kept pointing the same direction. Here they are, together, one more time.

# Where We'd Push Further

*Not weaknesses to apologize for — the honest boundary of what this analysis can claim*

**1**
Conversion rate (83.8%) is **20–80× higher** than typical baselines
EVIDENCE: Slide 6

**2**
cart_additions effect is **near-identical** across all 5 traffic sources
EVIDENCE: Slide 8

**3**
**Zero mutual correlation** among all 10 behavioral features
EVIDENCE: Technical Report, Section 4.4

**4**
A–B model gap **grows, not shrinks,** under a simpler algorithm
EVIDENCE: Slide 9

**5**
**100 sessions** share an identical, implausible outlier value (1000.0)
EVIDENCE: Slide 11

**The evidence is internally consistent.**
**The uncertainty is in the data — not the analytical method.**

Every technique here would apply unchanged to real ShopSphere data.
**What needs revisiting isn't the method — it's confirming the data matches what it's supposed to represent.**

**The method is ready. The next step is simple: validate the data — then act on what it reveals.**

# References & Resources

*Everything here is verifiable — nothing in this deck asks for trust without a way to check it*

## GitHub Repository

https://github.com/Bala-krsna/shopsphere-conversion-intelligence

Full notebook, cleaned code, and run history

Scan to open

## Built With

pandas | scikit-learn | SHAP | TensorFlow | Keras | scipy | matplotlib

All open-source; no proprietary tools required

## Full Citations & Acknowledgments

Complete literature review and reference list available in the Final Technical Report, Section 12

# Thank you